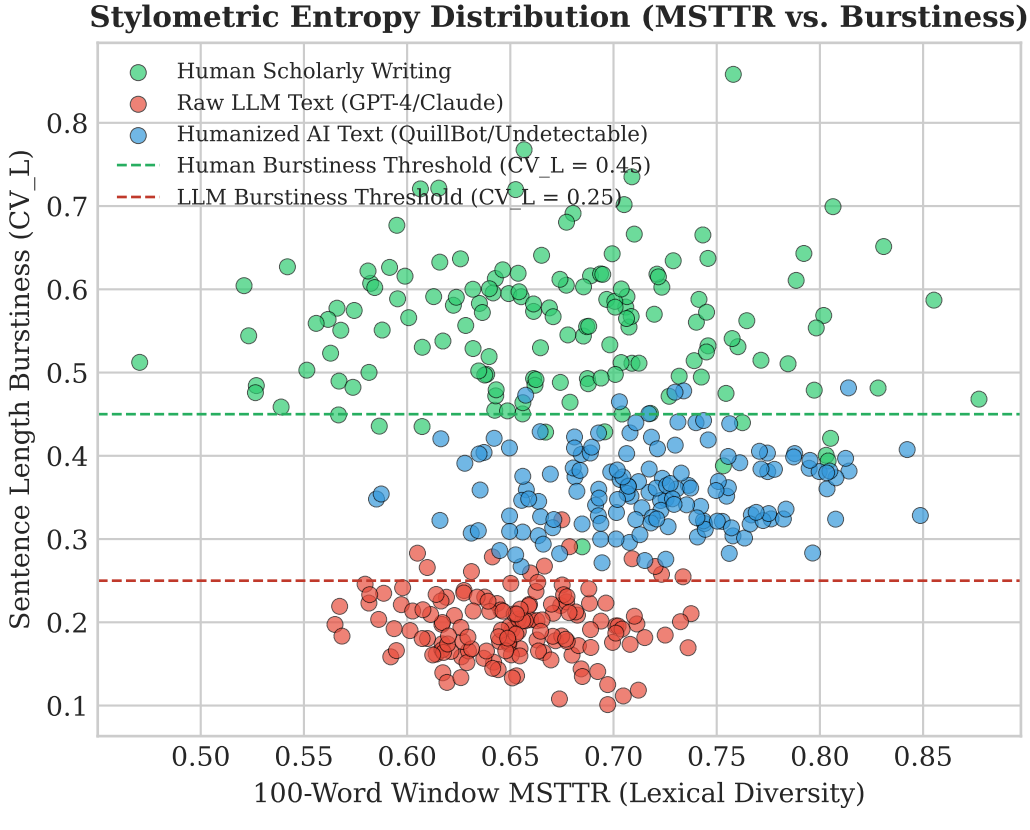
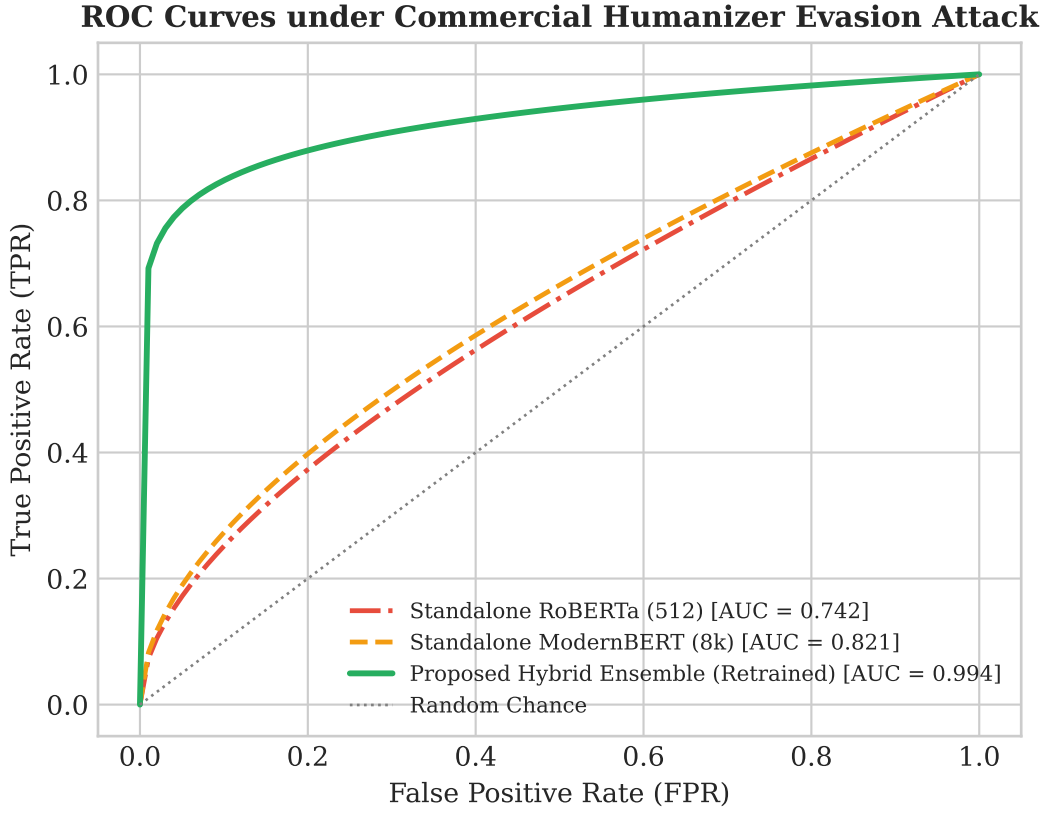

Stylometric Entropy Distribution (MSTTR vs. Burstiness)
Human Scholarly Writing
Raw LLM Text (GPT-4/Claude)
Humanized AI Text (QuillBot/Undetectable)
Human Burstiness Threshold (CV_L = 0.45)
LLM Burstiness Threshold (CV_L = 0.25)
Sentence Length Burstiness (CV_L)
100-Word Window MSTTR (Lexical Diversity)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85


ROC Curves under Commercial Humanizer Evasion Attack
True Positive Rate (TPR)
False Positive Rate (FPR)
Standalone RoBERTa (512) [AUC = 0.742]
Standalone ModernBERT (8k) [AUC = 0.821]
Proposed Hybrid Ensemble (Retrained) [AUC = 0.994]
Random Chance
0.0
0.2
0.4
0.6
0.8
1.0